

Hein Zarni Aung

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EDUCATION

Stony Brook University

B.S., Computer Science + Applied Math & Statistics (Data Science) | GPA: 3.40

Honors & Scholarships: Global Excellence Scholarship

Relevant Coursework: Full Stack Development, Systems Fundamentals I & II, Algorithms

Expected Graduation: May 2027

INTERNSHIPS

Software Engineering Intern — Clustr

Cross-platform React Native social/scheduling app | ~8-person dev team

Jan 2026 - Sept 2026 (expected)

- Built server-side community infrastructure using Firebase Cloud Functions, including CRUD operations for communities, posts, and comments with enforced ban/edit permissions to prevent client-side database tampering, plus join/leave workflows, like/dislike interactions, and activity feed endpoints.
- Designed and implemented a Scheduler UI module featuring tabbed navigation (Suggestions, Requests, Responses), a reschedule bottom sheet, custom date/time picker integration, and a calendar-completion animation to improve user engagement.
- Implemented secure authentication features, including a password reset flow (client screen plus backend cloud function) and custom claims to drive new-user onboarding logic.

Software Engineering Intern — Prenora Dynamics LLC

~8-person team

Jun 2026 - Sept 2026 (expected)

- Own end-to-end design decisions for an in-house, in-platform video chat feature, tasked with researching and evaluating available SDKs/approaches to replace reliance on third-party platforms (e.g., Zoom, Google Meet).
- Set up the PostgreSQL/Supabase database and backend data models for Jamory, an in-house app connecting students with music tutors to schedule and host lessons in one platform, and built a seed-data workflow to streamline debugging and local development.

RESEARCH

Undergraduate Research Assistant — Ethos Lab, Stony Brook University

PI: Abisheka Pitumpe | Team: 2

Sep 2025 - Dec 2025

- Built a real-time pipeline using Selenium and Python to monitor newly issued SSL/TLS certificates, scrape HTML/CSS/JS/URL content, and flag fraudulent domains impersonating EZ-Pass tolling services via keyword- and heuristic-based detection logic.

CAMPUS INVOLVEMENT

Teaching Assistant — Data Structures & Algorithms, Stony Brook University

Aug 2025 - Dec 2025

- Supported a 240-student Data Structures & Algorithms course through weekly office hours and recitations, guiding students through asymptotic analysis, recursion, and core data structures (linked lists, trees, heaps, hash tables, graphs).
- Created original review materials adopted as the basis for subsequent semesters' review sessions, and hosted 2-hour midterm review sessions reaching 100+ students combined in-person and via Zoom.

PROJECTS

TuneAcademy (React, NumPy, Essentia, Firestore)

- Built the AI/audio-analysis pipeline for an app matching music students with instructors based on specific technical weaknesses, using Essentia and NumPy to extract pitch, rhythm, and tone features from recorded performances.
- Set up a CI/CD pipeline (GitHub Actions) to lint, test, and auto-deploy to Cloudflare Workers, alongside a full-stack build spanning a React/Tailwind frontend, Python/FastAPI backend in Docker, and GCP-hosted Cloud Firestore.

SKILLS

Languages: TypeScript, JavaScript, Python, Java, C

Frontend: React, React Native, Next.js

Backend: Node.js, Express.js, FastAPI

Databases & Cloud: PostgreSQL, Firebase/Firestore, MongoDB, AWS, GCP

Tools: Git, Docker, Linux, PM2, Selenium